

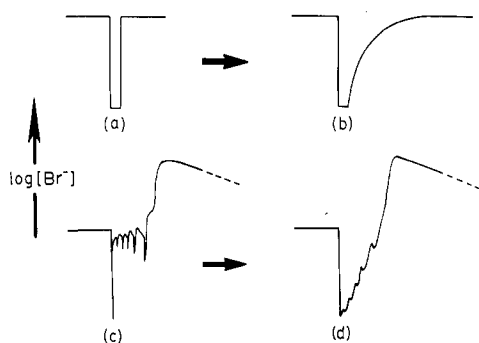


Figure 8. (a) Idealized behavior of a Br⁻-removing pulse. (b) Behavior when pulse in (a) is recorded by a Br⁻-detecting device with proposed delay. (c) Concentration-time profile of Br⁻ concentration in Oregonator model when Ag⁺ ions are added (see also Figure 6B). (d) Record of Br⁻ concentration shown in (c) by using a Br⁻-detecting device with proposed delay (see also Figure 6A).

turbing the BZ reaction by AgNO₃. While the platinum electrode, in agreement with presented calculations, shows high-frequency oscillations, the Br⁻-detecting electrode shows sometimes,^{11,21} but not always,¹⁹ a monotonic stepwise increase of the potential when the BZ reaction is treated with AgNO₃. Figure 6 illustrates this discrepancy. It is the absence of clear oscillations at the Br⁻-detecting electrode which makes Noszticzius and Ganapathisubramanian and Noyes postulate alternative mechanisms of control.^{11,16,17}

However, there seems to exist possible explanations other than changing the entire control mechanism. An alternative explanation of the behavior of the Br⁻-detecting electrode is qualitatively

indicated in Figure 8. The proposal is that when the Br⁻ concentration in BZ systems is driven to very low values, the Br⁻-detecting electrode may show a delay in its response. Figure 8a shows the idealized situation when Br⁻ ions are suddenly removed and then suddenly added again, while Figure 8b shows the (also idealized) corresponding situation when the Br⁻ concentration is now recorded by a Br⁻-detecting device with proposed delay. Figure 8c,d illustrates how the experimental potential of the Br⁻-detecting electrode in Ag⁺-perturbed BZ systems (see Figure 6A) can now be explained by such a delay mechanism.

We intend to test such assumptions critically by further experimental and theoretical work.

Conclusion

Although the Oregonator is only a skeleton of the FKN mechanism, the model reproduces qualitatively or semiquantitatively experimentally observed excitation thresholds, single and multiple excitation spikes, as well as period lengths in systems where AgNO₃ is pumped into the reaction vessel. Due to the simplicity of the model, there are second-order effects the model cannot account for, such as the frequencies at low Ag⁺ flow rates and the nonoscillatory behavior of Br⁻-detecting electrodes when the Br⁻ concentration by addition of AgNO₃ is forced to very low values.

However, besides these discrepancies, it seems fair to say that our results show that Ag⁺-induced oscillations and excitation phenomena can be explained and modeled within the framework of the FKN theory. Thus, there is presently no need to postulate other control mechanisms as done by Noszticzius¹³⁻¹⁶ and by Ganapathisubramanian and Noyes.¹⁷

Registry No. Ag, 7440-22-4; BrO₃⁻, 15541-45-4; Ce, 7440-45-1.

Negative Activation Energies and Curved Arrhenius Plots. 1. Theory of Reactions over Potential Wells

Michael Mozurkewich and Sidney W. Benson*

Donald P. and Katherine B. Loker Hydrocarbon Research Institute, University of Southern California, Los Angeles, California 90089 (Received: March 22, 1984)

The occurrence of negative activation energies and strongly curved Arrhenius plots in apparently elementary, gas-phase reactions is explained as being due to the formation of an intermediate complex. Activation energies as negative as -1.2 to -1.8 kcal mol⁻¹, for reactions near 300 K, may be explained by this mechanism. At higher temperatures, the activation energies should become increasingly positive. We present here a general procedure, based on RRKM theory, for calculating the rate constants of reactions of this kind.

I. Introduction

In recent years a number of gas-phase reactions which were believed to be elementary have been shown to have either strongly curved Arrhenius plots or strongly negative activation energies. For example, the reaction of hydroxyl radical with carbon monoxide has a strongly curved Arrhenius plot; the activation energy at 300 K is near zero, but at 2000 K it is about 7 kcal mol⁻¹.^{1,2} The reactions of hydroxyl radical with nitric acid³ and peroxyxynitric acid⁴ and the disproportionation of hydroperoxy radicals⁵ have negative activation energies of about -1.5 kcal mol⁻¹. In this series of papers we seek to explain these effects.

Dryer et al.⁵ have explained the curvature of the Arrhenius plot for the reaction OH + CO as being due to a tight transition state with a low threshold energy. Smith and Zellner⁶ interpreted this in terms of a stable intermediate complex. Margitan and Watson⁷ have proposed that a similar model may explain the negative temperature dependence for the reaction OH + HNO₃. Patrick, Barker, and Golden⁸ and Kircher and Sander⁹ have carried out calculations of the negative temperature dependence of the self-reaction of HO₂. In section II of this paper we present physical arguments to explain how the formation of a stable intermediate can produce negative action energies and curved Arrhenius plots. In section III we develop quantitative expressions, based on RRKM theory, that may be used to calculate the rates of these

(1) Baulch, D. L.; Drysdale, D. D. *Combust. Flame* **1974**, *23*, 215.

(2) Smith, I. W. M. *Chem. Phys. Lett.* **1977**, *49*, 112.

(3) DeMore, W. B.; Molina, M. J.; Watson, R. T.; Golden, D. M.; Hampson, R. F.; Kurylo, M. J.; Howard, C. J.; Ravishankara, A. R. Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, 1983, JPL Publ. 83-62.

(4) Smith, C. A.; Molina, L. T.; Lamb, J. J.; Molina, M. J. *Int. J. Chem. Kinet.* **1984**, *16*, 41.

(5) Dryer, F.; Nageli, D.; Glassman, I. *Combust. Flame* **1970**, *71*, 270.

(6) Smith, I. W. M.; Zellner, R. J. *Chem. Soc., Faraday Trans. 2* **1973**, *69*, 1617.

(7) Margitan, J. J.; Watson, R. T. *J. Phys. Chem.* **1982**, *86*, 3819.

(8) Patrick, R.; Barker, J. R.; Golden, D. M. *J. Phys. Chem.* **1984**, *88*, 128.

(9) Kircher, C. C.; Sander, S. P. *J. Phys. Chem.* **1984**, *88*, 2082.

reactions as a function of temperature. A simple method of estimating the pressure dependence of these reactions is given in section IV. In the following papers we apply this theory to the reactions of hydroxy radical with carbon monoxide, nitric acid, and peroxyxynitric acid.

The rate expressions developed here are formally equivalent to those used in chemical activation studies.¹⁰ It is possible to apply the chemical activation approach to reactions of the type considered in this series.^{8,9,11-13} However, chemical activation studies are usually concerned with the case where the intermediate does not undergo back-reaction; they focus on the branching ratio between dissociation and collisional deactivation. In the types of reactions considered here, the intermediate preferentially dissociates to reactants rather than products. For calculating the rates of these reactions the chemical activation formalism is unwieldy. The method presented here is more direct, especially in the manner in which angular momentum conservation is treated. Also, this method provides significant computational advantages.

II. Physical Basis for Negative Activation Energies and Curved Arrhenius Plots

In order to understand the origin of negative activation energies, we must first consider the Tolman interpretation¹⁴ of the activation energy. This defines the activation energy, E_{act} , in terms of the slope of an Arrhenius plot of the rate constant, k , vs. $1/T$. At any given temperature, T

$$E_{\text{act}} = -R(d \ln k / d(1/T)) = \langle E \rangle_{\text{TS}} - \langle E \rangle_{\text{R}} \quad (1)$$

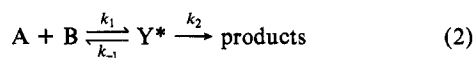
where $\langle E \rangle_{\text{TS}}$ is the average energy of the transition state and $\langle E \rangle_{\text{R}}$ is the average energy of the reactants. (Usually, the Tolman interpretation is given in terms of the average energy of those reactants which actually react; this is equal to the average energy of the transition state under the usual assumptions of RRKM theory.) Thus, the activation energy will be negative if reactants with low energy react faster than those with high energy.

The kinetic energy of the reactants will be essentially that due to the classical degrees of freedom, i.e., rotations and translations. For a bimolecular reaction some external degrees of freedom are lost as reactants proceed along the potential energy surface. If the transition state is loose, these will be replaced by internal rotations with about the same contribution to the average kinetic energy. Also, the potential energies will be essentially unchanged in forming the transition state. As a result, we expect to find $\langle E \rangle_{\text{TS}} \simeq \langle E \rangle_{\text{R}}$ and $E_{\text{act}} \simeq 0$.

If the reaction proceeds via a tight transition state, some classical degrees of freedom will be replaced with vibrational modes. These, if stiff, will contribute very little to the average internal energy; as a result, the average kinetic energy will decrease. However, such a transition state generally requires the breaking and forming of bonds with a substantial increase in potential energy. This will generally be much larger than the decrease in kinetic energy. Hence, we expect to find $E_{\text{act}} > 0$.

In order to obtain $E_{\text{act}} < 0$, we must have a tight transition state with a very low potential energy. We can achieve this if the reaction proceeds via a stable intermediate. A representative potential energy surface is shown in Figure 1.

The reaction scheme can be described by



If the second transition state (TS2) is tight and the first (TS1) is loose, k_2 will be much less than k_{-1} and the decomposition of Y^* to products will be rate determining. The tight transition state

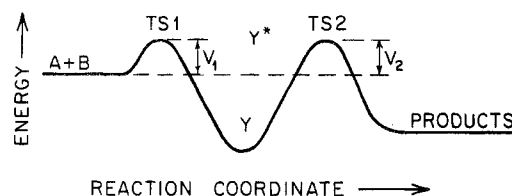


Figure 1. Potential energy surface for a bimolecular reaction proceeding via a stable intermediate.

for this step will have a significant threshold energy relative to the intermediate but may have a small, or even negative, potential energy relative to the reactants. If V_2 is sufficiently negative, then the second step will be faster than the first in spite of TS2 being tighter; this case will not concern us here. The reduction in the number of classical degrees of freedom can then produce a negative activation energy for the overall process.

At this point we must stress the fact that we are considering the case where the intermediate is sufficiently short-lived (or the pressure is sufficiently low) that it does not undergo any collisions. In the opposite extreme, where collisions are very frequent, the intermediate may be in equilibrium with the reactants. The rate will then be the product of an equilibrium constant and a rate constant. This will result in the transition-state theory (TST) rate constant; i.e., the rate constant will be expressed in terms of the equilibrium between the second transition state and the reactants. The activation energy in this rate expression will be simply the potential energy of the transition state relative to the reactants; this may be very negative. Large negative temperature dependences are in fact found in very fast solution reactions such as nitration. Although the following derivations are limited to the collision-free case, we will have occasion to compare the results to the TST limit.

As noted above, a negative activation energy implies that low-energy reactants react faster than high-energy ones. We can understand this by means of the following argument.¹⁵ For a given energy, E , the rate constant for formation of the intermediate will be $k_1(E)$. The fraction of the intermediate that will go on to products will be determined by the branching ratio, $\beta(E) = k_2(E)/k_{-1}(E)$. Given some reactant energy distribution, $f(E)$, the distribution, $g(E)$, for the second transition state is given by the proportionality (assuming steady state and neglecting collisions):

$$g(E) = \frac{k_1(E) k_2(E) f(E)}{k_{-1}(E) + k_2(E)} = k_1(E) \beta(E) f(E) / (1 + \beta(E))$$

If the first transition state is loose and the second is tight, then $k_1(E)$ will be essentially constant and $\beta(E)$ will decrease with increasing E . As a result, $g(E)$ will be shifted to lower energies than $f(E)$; i.e., the reaction will be faster at lower energy. This is due not to the individual rate constants, $k_1(E)$ or $k_2(E)$, being larger at small E but to the fact that once the intermediate is formed at low E , it returns to reactants relatively slowly. As a result, it is more likely to go on to products.

An alternate way of looking at this is to note that the density of states increases more slowly for the intermediate than for the reactants. As a result, the microcanonical equilibrium constant, $K_{\text{eq}}(E) = k_1(E)/k_{-1}(E)$, decreases with increasing E . This lowers the average energy of the intermediate and, therefore, of the second transition state. That the intermediate has a lower average energy than the reactants can also be seen by applying the arguments of Appendix A to the intermediate.

In order to determine how negative the activation energy might be, we need to determine $\langle E \rangle_{\text{TS}}$. In Appendix A we estimate this quantity and determine the activation energy for various limiting cases. We find that if the potential energy of the transition state is sufficiently negative, then the activation energy approaches $RT - \langle E \rangle_{\text{R}}$. If the reactants have a large number of classical degrees of freedom, this can be very negative. Also, we show that if the transition state has many degrees of freedom, then the activation

(10) P. J. Robinson and K. A. Holbrook, "Unimolecular Reactions"; Wiley: New York, 1972.

(11) Olmstead, W. N.; Brauman, J. I. *J. Am. Chem. Soc.* **1977**, *99*, 4219.

(12) Chesnavich, W. J.; Bass, L.; Su, T.; Bowers, M. T. *J. Chem. Phys.* **1981**, *74*, 2228.

(13) Bass, L. M.; Cates, R. D.; Jarrold, M. F.; Kircher, N. J.; Bowers, M. T. *J. Am. Chem. Soc.* **1983**, *105*, 7024.

(14) Levine, R. D.; Bernstein, R. B. "Molecular Reaction Dynamics"; Oxford University Press: New York, 1974.

(15) Golden, D. M. *J. Phys. Chem.* **1979**, *83*, 108.

energy approaches $V_{TS} - mRT/2$, where m is the change in the number of phase space degrees of freedom which contribute $RT/2$ to the average energy (i.e., for rotations $m = 1$ and for classical vibrations $m = 2$). This is simply the transition-state theory (high pressure) activation energy. In this case we have enough degrees of freedom that the reactants, in effect, carry their own heat bath along with them.

The third limit is one in which we have few degrees of freedom or a transition-state potential energy that is only slightly negative. In this case the activation energy approaches $-mRT/2$. This is just what it would be if $V_2 = 0$.

In the subsequent papers of this series we present detailed calculations on systems in which the reactants are small molecules, generally with no internal rotations. As a result, the classical degrees of freedom in the reactants are three translations (the center of mass motion is factored out) and four to six rotations. The transition state will have three rotations plus the reaction coordinate, which contributes RT to the average energy. Hence m , the number of classical phase space degrees of freedom lost, will be between two and four.

Therefore, when $V_2 < 0$, the activation energy should fall in the range

$$-RT > E_{\text{act}} > -5RT/2 \quad \text{for linear reactants}$$

$$-2RT > E_{\text{act}} > -7RT/2 \quad \text{for nonlinear reactants}$$

For the gas-phase reactions of hydroxyl radical with nitric and peroxy acids and the disproportionation of HO_2 , the observed activation energies are about $(-5/2)RT$. This is in excellent agreement with the above limits.

So far we have only considered the active modes of the complex, i.e., those degrees of freedom which may randomly exchange energy with the reaction coordinate. These active modes serve as a heat bath. Since the formation of the transition state is exothermic, the flow of energy is, on average, from the reaction coordinate to the active modes. This flow is the opposite of what occurs in unimolecular decomposition.

We must also consider adiabatic modes of the complex. These are modes whose quantum states are fixed by some conservation condition. Conservation of angular momentum requires that some rotational degrees of freedom of the complex be adiabatic. The energy associated with these rotations can change only if the moments of inertia, and, therefore, the energy level spacing, change. Hence, these degrees of freedom constitute work reservoirs. Since the moments of inertia decrease in going from the first, loose transition state to the second, tight transition state, energy will also flow from the reaction coordinate into these modes. Consequently, the adiabatic degrees of freedom will have the same qualitative effect as the active modes although they may have somewhat different quantitative effects.

If k_{-1} is large compared to k_2 , then the intermediate will be essentially in equilibrium with the reactants. In this case we will not be able to tell if the angular momentum, J , was conserved in forming the intermediate. Since there should be little change in angular momentum in going from the intermediate to the second transition state, the same random J distribution should appear in the second transition state. Consequently, in this case, we expect that there will be very little difference between treating the rotations as active or adiabatic modes.

If k_{-1} is not large compared to k_2 , then the branching ratio, k_2/k_{-1} , will have a significant effect on the rate. Since the two transition states will have substantially different rotational constants, the branching ratio will depend strongly on J . In this case we expect the adiabatic modes to behave quite differently from the active modes.

Finally, we note that in RRKM calculations on unimolecular decomposition it is common to account for adiabatic rotations by using the approximation of Waage and Rabinovitch.¹⁶ If we regard k_1 and k_{-1} as replacing the collisional activation and deactivation, we have, when $k_{-1} \gg k_2$, a situation that is anal-

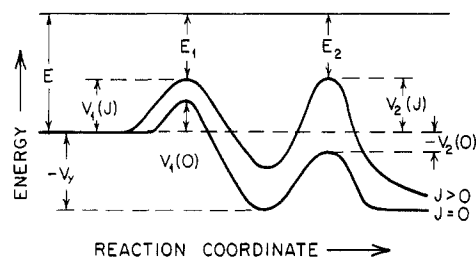


Figure 2. Illustration of energy terminology. The potential curve labeled $J > 0$ includes the rotational energy.

ogous to the high-pressure limit. In this case the Waage-Rabinovitch approximation amounts to including in the rate expression a factor consisting of the partition function for the adiabatic rotations. For the systems considered here this approximation will not hold. In effect, it assumes that there will be rotational energy RT in addition to the energy associated with the other degrees of freedom. However, as we mentioned above and shown in Appendix A, when the potential energy of the transition state is sufficiently negative, the total internal energy is RT . This is independent of the number of degrees of freedom. Hence, the adiabatic modes do not appear to contribute to the average energy. This does not imply that there is no energy in these modes; it only implies that this energy comes at the expense of other degrees of freedom.

III. Derivation of Rate Expressions

In order to derive a rate expression for the reaction scheme in eq 2, we must first consider Figure 2 which is an elaboration of Figure 1. The potential energies are measured relative to the ground state of the reactants, A and B. $V_1(0)$ and $V_2(0)$ are the potential energies of the transition states for the nonrotating ($J = 0$) complex. If the first transition state is loose, then $V_1(0) = 0$. $V_2(0)$ may be positive or negative. The ground state of the intermediate is at a negative energy, V_0 , relative to the reactants.

If the intermediate is rotating, then conservation of angular momentum requires that there be some minimum kinetic energy (Appendix B). We add this energy to the potential energy to get an effective potential:

$$V_1(J) = V_1(0) + B_1 J(J+1) \quad (3)$$

$$V_2(J) = V_2(0) + B_2 J(J+1) \quad (4)$$

Here B_1 and B_2 are the rotational constants for the two transition states; for simplicity we assume that the geometries are independent of J . We also assume that the first transition state is looser, i.e., has a larger moment of inertia, so that $B_2 > B_1$. The larger of $V_1(J)$ or $V_2(J)$ determines the minimum total energy, E_0 , needed for reaction to take place. We will call this quantity $V_0(J)$.

The difference between the total energy and the rotational potential provides the energies E_1 and E_2 available for distribution among internal modes of the transition states. These quantities are given by

$$E_1 = E - V_1(J) = E - (V_1(0) + B_1 J(J+1)) \quad (5)$$

$$E_2 = E - V_2(J) = E - (V_2(0) + B_2 J(J+1)) \quad (6)$$

If the intermediate were in complete equilibrium with the reactants, then the rate could be obtained by applying transition-state theory to the second transition state. The actual rate will differ from this for two reasons. First, there will be no population of the second transition state with E_2 such that

$$E_2 + V_2(J) < V_1(J) \quad (7)$$

i.e., with $E < V_0(J)$. As a result, for any given $E_2 < V_1(0) - V_2(0)$ there will be no contribution to the rate for J less than some minimum value. The second correction is due to the fact that the first step in eq 2 takes place at a finite rate. As a result, the concentration of the intermediate will be determined by steady state rather than equilibrium conditions.

Both of these corrections will usually be fairly small, the first because it applies only to a small range of energies and the second because k_{-1} will generally be larger than k_2 so that the steady-state concentration will not be much different from the equilibrium concentration. It is therefore computationally convenient to calculate the TST rate and the corrections separately and then combine them into the final result. In the derivation that follows we derive a general expression which we then split up into these three terms.

Referring to eq 2, we note that k_2 and the concentration of the intermediate will be functions of the total energy, E , and the angular momentum, J . The reaction rate is given by

$$-d[A]/dt = k_{\text{exptl}}[A][B] = \int_{V_0(0)}^{\infty} dE \sum_{J=0}^{J_M} k_2(E, J) [Y(E, J)] \quad (8)$$

where k_{exptl} is the experimental rate constant. J_M is the maximum angular momentum for which reaction can take place with energy E . It is determined by the condition

$$E = V_0(J_M)$$

$V_0(J_M)$ is the larger of $V_1(J_M)$ (eq 3) or $V_2(J_M)$ (eq 4).

The equilibrium concentration of the intermediate, with energy in the range E to $E + dE$ and with angular momentum J , is given by

$$[Y(E, J)]_{\text{eq}} = N_Y(E, J) \exp((V_Y - E)/RT) [Y]_{\text{tot}} / Q_Y \quad (9)$$

where $N_Y(E, J)$ is the density of states, $[Y]_{\text{tot}}$ is the total equilibrium concentration of Y , and Q_Y is the partition function for Y with the center of mass motion factored out. We assume that the intermediate has just one electronic state. Any degeneracy, such as spin or reaction path degeneracy, is included in $N_Y(E, J)$. The steady-state concentration of the intermediate is then

$$[Y(E, J)] = [Y(E, J)]_{\text{eq}} / (1 + \beta(E, J)) \quad (10)$$

where the branching ratio, $\beta(E, J)$, is given by

$$\beta(E, J) = k_2(E, J) / k_{-1}(E, J) \quad (11)$$

Substituting (9) and (10) into (8) and noting that

$$[Y]_{\text{tot}}/[A][B] = K_{\text{eq}} = Q_Y \exp(-V_Y/RT) / Q_A Q_B$$

we obtain for the experimental rate constant

$$k_{\text{exptl}} = \frac{1}{Q_A Q_B} \int_{V_0(0)}^{\infty} dE \sum_{J=0}^{J_M} \frac{k_2(E, J) N \exp(-E/RT)}{1 + \beta(E, J)} \quad (12)$$

The center of mass partition function is assumed to have been factored out of the product $Q_A Q_B$.

We may use RRKM theory to obtain expressions for $k_2(E, J)$ and $k_{-1}(E, J)$. Let $N_2(E^+)$ be the number of states of the activated complex with energy E^+ available for redistribution among the active degrees of freedom. The sum of states, $W_2(E_2)$, with energy less than or equal to E_2 (eq 6) is

$$W_2(E_2) = \sum_{E^+=0}^{E_2} N_2(E^+) \quad (13)$$

Each of these states has a degeneracy, due to adiabatic rotations, of $(2J + 1)^{S_2}$, where S_2 is 1 or 2 depending on whether the activated complex is nonspherical or spherical (see Appendix B). We can now obtain the expression for k_2 :¹⁰

$$k_2(E, J) = \frac{(2J + 1)^{S_2} W_2(E_2)}{h N_Y(E, J)} \quad (14)$$

Again, we have included any degeneracy, such as spin, in N_Y .

A similar equation to (14) holds for the decomposition to reactants:

$$k_{-1}(E, J) = \frac{(2J + 1)^{S_1} W_1(E_1)}{h N_Y(E, J)} \quad (15)$$

Substituting (14) and (15) into (11), we obtain for the branching ratio

$$\beta(E, J) = (2J + 1)^{(S_1 - S_2)} W_2(E_2) / W_1(E_1) \quad (16)$$

Substituting (14) into (12), we have for the rate constant

$$k_{\text{exptl}} = \frac{1}{h Q_A Q_B} \int_{V_0(0)}^{\infty} dE \sum_{J=0}^{J_M} \frac{(2J + 1)^{S_2} W_2(E_2) \exp(-E/RT)}{1 + \beta(E, J)} \quad (17)$$

In order to put this expression into a more convenient form, we replace the integration over E with one over E_2 . In the resulting integral E_2 can take any nonnegative value. For a given E_2 the condition $E \geq V_1(J)$ must be satisfied by J , i.e.

$$V_2(J) + E_2 \geq V_1(J)$$

which, by using (3) and (4), may be rearranged to

$$J(J + 1) \geq ((V_1(0) - V_2(0)) - E_2) / (B_2 - B_1) \quad (18)$$

Also, we must have $J \geq 0$. We will let J_0 designate the minimum value of J which satisfies both of these conditions. If $V_2(0) > V_1(0)$ or if E_2 is sufficiently large, we will have $J_0 = 0$.

Using these limits and eq 6, we may transform eq 17 to

$$k_{\text{exptl}} = Z \int_0^{\infty} dE_2 W_2(E_2) \exp(-E_2/RT) \times \sum_{J=J_0}^{\infty} \frac{(2J + 1)^{S_2} \exp(-B_2 J(J + 1)/RT)}{1 + \beta(E_2, J)} \quad (19)$$

where

$$Z = \frac{\exp(-V_2(0)/RT)}{h Q_A Q_B}$$

As we will see below, if $J_0 = 0$ and $\beta = 0$, this reduces to the transition-state theory expression. This suggests that we make the substitutions

$$1/(1 + \beta) = 1 - \beta/(1 + \beta)$$

and

$$\sum_{J=J_0}^{\infty} = \sum_{J=0}^{\infty} - \sum_{J=0}^{J_0}$$

in order to split (19) into three terms:

$$k_{\text{exptl}} = k_{\text{TST}} - C_J - C_B \quad (20)$$

The first term in (20) is the TST rate constant. This is given by

$$k_{\text{TST}} = Z \int_0^{\infty} dE_2 W_2(E_2) \exp(-E_2/RT) \sum_{J=0}^{\infty} (2J + 1)^{S_2} \exp(-B_2 J(J + 1)/RT) \quad (21)$$

The sum over J is simply the partition function for the adiabatic rotations of the transition state. The evaluation of the integral over E_2 is standard;¹ it is RT times the partition function for the active degrees of freedom. Hence, (21) becomes

$$k_{\text{TST}} = (RT Q_2 / h Q_A Q_B) \exp(-V_2(0)/RT)$$

which is simply the TST rate constant.

The second term in (20) is a correction for the fact that the low rotational states may not be occupied. It is

$$C_J = Z \int_0^{V_1(0) - V_2(0)} dE_2 W_2(E_2) \exp(-E_2/RT) \sum_{J=0}^{J_0} (2J + 1)^{S_2} \exp(-B_2 J(J + 1)/RT)$$

The upper limit on E_2 is determined by the point above which $J_0 = 0$ (see eq 18). We can write the sum over J as an integral and let

$$\alpha = J(J + 1)$$

to get

$$C_J = Z \int_0^{V_1(0) - V_2(0)} dE_2 W_2(E_2) \exp(-E_2/RT) \int_0^{\alpha_0} d\alpha (1 + 4\alpha)^{(S_2 - 1)/2} \exp(-B_2 \alpha / RT) \quad (22)$$

The third term in (20) is a correction for the fact that the branching ratio, β , is not zero. It is

$$C = Z \int_0^\infty dE_2 W_2(E_2) \exp(-E_2/RT) \sum_{J=J_0}^\infty [(2J+1)^{S_2} \beta(E_2, J) \exp(-B_2 J(J+1)/RT)] / [1 + \beta(E_2, J)]$$

Changing the sum to an integral and letting

$$X_0 = B_2 J_0(J_0 + 1)/RT$$

and

$$X = [B_2(J(J+1)/RT)] - X_0$$

this becomes

$$C = (RTZ/B_2) \int_0^\infty dE_2 W_2(E_2) \exp(-X_0 - E_2/RT) \int_0^\infty dX [1 + 4RT(X + X_0)/B_2]^{(S_2-1)/2} (e^{-X}) \beta(E_2, X) / (1 + \beta(E_2, X)) \quad (23)$$

where

$$\beta(E_2, X) = (2J+1)^{(S_2-S_1)/2} W_2(E_2) / W_1(E_1) \quad (24)$$

and

$$2J+1 = [1 + 4RT(X + X_0)/B_2]^{1/2}$$

and

$$E_1 = E_2 + (V_2(0) - V_1(0)) + (B_2 - B_1)RT(X + X_0)/B_2$$

Putting the equations in this form is convenient for numerical evaluation. The largest term, k_{TST} , can be evaluated directly. The two corrections both require double integrations, but both can be done quickly. In evaluating C_J (eq 22), one need consider only a limited range of values of E_2 and α . The integrations in eq 23 generally cover rapidly.

As X increases, E_1 , and therefore $W_1(E_1)$, gets larger and $\beta(E_2, X)$ (eq 24) decreases. Therefore, the inner integrand in (23) rapidly gets very small so that the integral will converge quickly. Also, as E_2 increases, E_1 increases. If the first transition state is looser than the second, this will cause W_1 to increase faster than W_2 and β will decrease. As a result, the outer integral will converge at reasonably small values of E_2 .

In some cases we may find that these corrections are not small compared to the TST rate. In these cases, to prevent large numerical errors, we must carry out a direct numerical integration of eq 19. This can be very time consuming, especially at high temperatures where the integral may not converge until E_2 is many tens of kcal mol⁻¹. Fortunately, as the temperature increases, the corrections to the TST rate become smaller so the two methods compliment each other.

IV. Pressure Dependence

Reactions proceeding via an excited intermediate should exhibit a pressure dependence due to collisional deexcitation of the intermediate. For the reactions considered in the subsequent papers of this series, the observed pressure-dependent portion of the rate is, at most, comparable to the low-pressure limit. The rate of formation of the excited intermediate is much faster than this. As a result, it should be a good approximation to treat the pressure-independent and -dependent portions of the rate as additive.

We describe the pressure dependence in terms of a third-order rate constant, k_{III} , for the formation of stabilized intermediate. We express this as

$$k_{\text{III}} = \lambda Z_{\text{coll}} K_{\text{eq}}^* \quad (25)$$

where λ is the collision efficiency, Z_{coll} is the collision rate constant, and K_{eq}^* is the equilibrium constant between the excited intermediate and the reactants.

In order to calculate K_{eq}^* , we assume that states of the intermediate which are accessible, i.e., which may be formed from the reactants while conserving energy and angular momentum, have the same populations that they would have at equilibrium. States which are inaccessible are assumed to have zero population.

Then we can express K_{eq}^* in terms of the partition functions of the excited intermediate, Y^* , and reactants, $A + B$:

$$K_{\text{eq}}^* = Q_Y^* / Q_A Q_B \quad (26)$$

Q_Y^* is obtained by summing, with appropriate Boltzmann factors, over those states of Y^* which are accessible. The center of mass motion is assumed to be factored out of both Q_Y^* and $Q_A Q_B$.

The intermediate will have rotational energy

$$E_{\text{rot}} = B_Y J(J+1) \quad (27)$$

where B_Y is the rotational constant. This energy will not be available for random redistribution among the other degrees of freedom (active modes) of the intermediate. The energy at which the density of states, N_{act} , of the active modes is to be evaluated is then $E + V_Y - E_{\text{rot}}$, where E is the total energy (measured relative to the reactants) and V_Y is the energy of the ground state of the intermediate.

The partition function for the excited intermediate may now be written as

$$Q_Y^* = \int_{V_1}^\infty dE e^{-E/RT} \left[\sum_{J=0}^{J_{\text{max}}(E)} (2J+1) N_{\text{act}}(E + V_Y - E_{\text{rot}}) \right] \quad (28)$$

where $J_{\text{max}}(E)$ is the maximum angular momentum for which the intermediate may be formed with energy E . This must satisfy two conditions. First, the total energy, E , must be sufficient to clear the centrifugal barrier, $V_1 + B_1 J(J+1)$, at TS1. This requires that

$$E_{\text{rot}} \leq B_Y(E - V_1)/B_1 \quad (29)$$

Second, the nonfixed energy must be nonnegative, i.e.

$$E_{\text{rot}} \leq E - V_Y \quad (30)$$

If this latter condition is not satisfied, the intermediate will be unstable due to centrifugal forces.

We let E_c designate the energy at which condition (30) becomes more stringent than condition (29), i.e.

$$E_c = B_Y V_1 - B_1 V_Y / (B_Y - B_1) \quad (31)$$

If one replaces the summation over J with an integration over E_{rot} equation, (28) becomes

$$Q_Y^* = \int_{V_1}^{E_c} dE \exp\left(-\frac{E}{RT}\right) \int_0^{B_Y(E-V_1)/B_1} dE_{\text{rot}} \left(\frac{1}{B_Y}\right) N_{\text{act}}(E - V_Y - E_{\text{rot}}) + \int_{E_c}^\infty dE \exp\left(-\frac{E}{RT}\right) \int_0^{E-V_Y} dE_{\text{rot}} \left(\frac{1}{B_Y}\right) N_{\text{act}}(E - V_Y - E_{\text{rot}}) \quad (32)$$

There are two special cases of eq 32 which will be of interest to us. If the well is very deep and TS1 is not too loose (i.e., B_1 is not much larger than B_Y), then E_c becomes very large. We can then approximate (32) by only the first term:

$$Q_Y^* = \int_{V_1}^\infty dE \exp\left(-\frac{E}{RT}\right) \int_0^{B_Y(E-V_1)/B_1} dE_{\text{rot}} \left(\frac{1}{B_Y}\right) N_{\text{act}}(E - V_Y - E_{\text{rot}})$$

If the well is deep, then N_{act} will vary slowly with energy. We treat N_{act} as constant and evaluate it at $E = V_1 + E_{\text{rot}}$. After integrating, we obtain

$$Q_Y^* = (B_Y/B_1) RT Q_{\text{rot}}^{\text{ad}} N_{\text{act}}(V_1 - V_Y) \exp(-V_1/RT) \quad (33)$$

where $Q_{\text{rot}}^{\text{ad}} = RT/B_Y$ is the partition function for adiabatic rotations of the intermediate. $N(V - V_Y)$ is calculated by using the Whitten-Rabinovitch approximation.¹⁷

From eq 33 we see that the first transition state affects the partition function in two ways. The threshold energy, V_1 , limits the number of states which are energetically accessible.¹ However, if the moment of inertia decreases in going from the transition

(17) Forst, W. "Theory of Unimolecular Reactions"; Academic Press: New York, 1973.

state to the intermediate, then energy will flow into the rotations from the active modes and states with $E < V_1 + E_{\text{rot}}$ will be inaccessible. Similarly, if $B_T > B_Y$, some states with $E < V_1 + E_{\text{rot}}$ will be inaccessible. In this manner the ratio of the moment of inertia of the transition state to the moment of the intermediate will help determine the number of states counted in determining Q_Y^* .

The second special case of interest is that where TS1 is very loose, so that $B_Y \gg B_1$, and the well depth is not too great. In this case, E_c (eq 31) approaches V_1 and only the second term in (32) is important. Thus

$$Q_Y^* = \int_{V_1}^{\infty} dE \exp\left(-\frac{E}{RT}\right) \int_0^{E-V_1} dE_{\text{rot}} \left(\frac{1}{B_Y}\right) N_{\text{act}}(E - V_1 - E_{\text{rot}})$$

The inner integral is simply the convolution of the density of states for the adiabatic rotations and the active modes.¹⁷ We may therefore write

$$Q_Y^* = \int_{V_1}^{\infty} dE \exp\left(-\frac{E}{RT}\right) N_{\text{tot}}(E - V_1) \quad (34)$$

where N_{tot} is the density of states for all degrees of freedom of the intermediate. Since we are assuming a rapid equilibrium with essentially no centripetal barrier, there is nothing in eq 34 to distinguish adiabatic and active degrees of freedom.

V. Summary

For bimolecular gas-phase reactions, negative activation energies and strongly curved Arrhenius plots may be explained by assuming that an intermediate complex is formed. This permits the rate-determining step to have a tight transition state with a small or negative potential energy relative to the reactants. As a result, at low temperature the average energy of the transition state will be less than that of the reactants and the activation energy will be negative. For small reactants, near room temperature, the activation energy may be as negative as -1.5 to -2.1 kcal mol⁻¹. As the temperature increases, the vibrational energy of the transition state becomes significant and the Arrhenius plot curves upward.

We have derived an expression, based on RRKM theory, for the rate of this type of reaction. This expression may be used when the second transition state has either a positive or negative potential energy relative to the reactants. It also allows for the possibility that the first step may not be very fast compared to the second and explicitly accounts for the conservation of angular momentum. The rate may be evaluated either by direct integration or by calculating corrections to the transition-state theory rate. The latter procedure should require much less computation in most cases.

Acknowledgment. This work was supported by the National Science Foundation and the Army Research Office under Grants No. CHE-79-26623 and DAAG29-82-K-0043. Special thanks to Dr. John J. Lamb for the initial impetus to and the helpful discussions during this work.

Appendix A

In section II we noted that the average energy of the transition state determines the activation energy of a reaction. If a chemical reaction proceeds via a transition state with a positive threshold energy, the average internal energy of the transition state may be obtained in the same way as for a stable molecule. However, if the threshold energy is negative, as for the second transition state in Figure 1, states with a total energy less than zero must be excluded from the average. In this appendix we calculate this average. The average energy of the intermediate may also be calculated by the same method.

For simplicity, in the following we will exclude rotational effects and assume that all degrees of freedom may be treated classically. The average energy of the transition state is

$$\langle E \rangle_{\text{TS}} = -R \, d \ln Q_{\text{TS}} / d(1/T) + V_{\text{TS}} \quad (\text{A.1})$$

where V_{TS} is the (negative) potential energy of the ground state and Q_{TS} is the partition function. Letting the Hamiltonian, H , be measured relative to the threshold and requiring that the total energy, $E = H + V_{\text{TS}}$, be nonnegative, we have

$$Q_{\text{TS}} = \int_{H \geq -V_{\text{TS}}}^{\infty} \dots \int dq \dots dp \exp(-H/RT) \quad (\text{A.2})$$

Instead of integrating over all of phase space, we exclude that region where $E < 0$.

To evaluate this integral, we replace one of the p 's or q 's with H . The integral now becomes¹⁸

$$Q_{\text{TS}} = C \int_{-V_{\text{TS}}}^{\infty} dH H^{(n-2)/2} \exp(-H/RT) \quad (\text{A.3})$$

where n is the number of degrees of freedom. The constant C represents the integration over the $n-1$ variables other than H ; it is independent of H , V_{TS} , and T . Substituting (A.3) into (A.1), we obtain

$$\langle E \rangle_{\text{TS}} = \frac{(nRT/2) + RTC(-V_{\text{TS}})^{n/2} \exp(V_{\text{TS}}/RT) / Q_{\text{TS}} + V_{\text{TS}}}{(nRT/2) + RTC(-V_{\text{TS}})^{n/2} \exp(V_{\text{TS}}/RT) / Q_{\text{TS}} + V_{\text{TS}}} \quad (\text{A.4})$$

The first term in this expression is just the usual result from equipartition of energy. If we have $V_{\text{TS}} \geq 0$, then the lower limit of integration should be zero and the second term in (A.4) should not appear.

to make further progress, we must evaluate Q_{TS} . The integral in (A.3) can only be evaluated for even values of n . The conclusions we draw below should not depend on whether n is even or odd so we will restrict ourselves to even n . In this case (A.3) becomes

$$Q_{\text{TS}} = C [\exp(-V_{\text{TS}}/RT)] (-V_{\text{TS}})^{n/2} [(n-2)/2)! \sum_{i=0}^{(n-2)/2} [(RT/-V_{\text{TS}})^{i+1}] / [(((n-2)/2) - i)!] \quad (\text{A.5})$$

Substituting (A.5) into (A.4) and letting

$$X = -V_{\text{TS}}/RT$$

we get

$$\langle E \rangle_{\text{TS}} = RT[(n/2) - X + X^{n/2} / ((n-2)/2)! \sum_{j=0}^{(n-2)/2} (X^j/j!)]$$

which may be rearranged to

$$\langle E \rangle_{\text{TS}} = RT \left[\sum_{j=0}^{(n-2)/2} ((n/2) - j) X^j/j! \right] / \left[\sum_{j=0}^{(n-2)/2} X^j/j! \right] \quad (\text{A.6})$$

The reactants will have average energy

$$\langle E \rangle_R = (n+m)RT/2 \quad (\text{A.7})$$

where m is the number of degrees of freedom lost in forming the transition state. From (A.6), (A.7), and (1) we obtain, provided that $n \geq 4$

$$E_{\text{act}} = (-RTX) \left[\frac{\sum_{j=0}^{(n-4)/2} X^j/j!}{\sum_{j=0}^{(n-2)/2} X^j/j!} \right] - (mRT)/2 \quad (\text{A.8})$$

For small X we have

$$E_{\text{act}} \rightarrow -mRT/2 \quad \text{when } -V_{\text{TS}} \ll RT \quad (\text{A.9})$$

and for large X

$$E_{\text{act}} \rightarrow RT - \langle E \rangle_R \quad \text{when } -V_{\text{TS}} \gg nRT/2 \quad (\text{A.10})$$

Since there are two degrees of freedom in phase space associated with the reaction coordinate, we must have $n \geq 2$. When $n = 2$, eq A.8 is replaced by

(18) Courant, R. "Differential and Integral Calculus"; Interscience: New York, 1936.

$$E_{\text{act}} = -mRT/2 \quad (\text{A.11})$$

When n is sufficiently large for both summations to converge to the exponential, (A.8) becomes

$$E_{\text{act}} \rightarrow V_{\text{TS}} - mRT/2 \quad (\text{A.12})$$

This is the same activation energy that we get in the TST limit with many collisions. This is because the large number of classical degrees of freedom serves as a heat bath just as collision partners would at high density.

The limit in eq A.10 is the one of interest for significant negative activation energies in reactions between small molecules. It states that the average energy of the transition state is just RT so that the activation energy is just RT minus the internal energy of the reactants. This has a simple physical explanation. At energies well above the zero point the density of states changes slowly relative to the Boltzmann factor. As a result, the average energy is the same as for systems with a constant density of states, such as a classical oscillator or a two-dimensional rotor.

Appendix B

For a molecule with principal moments of inertia I_x , I_y , and I_z , the kinetic energy is

$$E_{\text{rot}} = (L_x^2/I_x) + (L_y^2/I_y) + (L_z^2/I_z)$$

and the total angular momentum, K , is determined by

$$L^2 = L_x^2 + L_y^2 + L_z^2$$

If we assume that $I_x \leq I_y \leq I_z$, then, for a particular value of L , the energy must lie in the range

$$(L^2/I_z) \leq E_{\text{rot}} \leq (L^2/I_x) \quad (\text{B.1})$$

An amount of energy equal to the lower limit is unavailable for exchange with other degrees of freedom. We take this into account by assuming that we have an adiabatic two-dimensional rotor with moment of inertia I_z . The energy associated with this rotor can change only through a change in geometry which alters I_z .

The total rotational energy can vary over the range indicated in eq B.1. If we have $I_z \gg I_x$, then the upper limit will be very large. In this case we may take the remaining degree of freedom as a one-dimensional rotor. The moment of inertia must be chosen so as to produce the correct overall partition function for the three rotational degrees of freedom.

If I_x is not very different from I_z , there will be very little energy available for exchange with other degrees of freedom. In this case we should treat the molecule as a spherical top with moments $(I_x I_y I_z)^{1/3}$. All three rotational degrees of freedom will be adiabatic. The degeneracy associated with J is $(2J+1)^2$ rather than $(2J+1)$ as in the case where $I_z \gg I_x$.

Negative Activation Energies and Curved Arrhenius Plots. 2. OH + CO

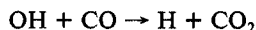
Michael Mozurkewich, John J. Lamb, and Sidney W. Benson*

Department of Chemistry, Donald P. and Katherine B. Loker Hydrocarbon Research Institute, University of Southern California, Los Angeles, California 90089 (Received: March 22, 1984)

In the first paper of this series we have presented a method for calculating the rate constants of reactions proceeding through an intermediate complex. Here we apply that method to the reactions OH + CO and OD + CO over a temperature range of 200–2000 K. A number of models of the transition state for decomposition of the intermediate to products were examined. The best results were achieved by using a bent transition state with the OCO bend as reaction coordinate. The pressure dependence of these reactions can be explained only if the transition state for formation of the intermediate is fairly tight with a significant threshold energy.

I. Introduction

The gas-phase reaction of hydroxyl radical with carbon monoxide



is an important component of the atmospheric HO_x cycle. It also plays a critical role in combustion processes. There are two aspects of this reaction which suggest that it is not a simple elementary reaction. First, the Arrhenius plot is strongly curved;¹ the activation energy is near zero at 300 K and increases to about 7 kcal mol⁻¹ at 2000 K. The second interesting aspect is that the rate constant increases with pressure.²⁻⁵

Dryer et al.⁶ showed that the temperature dependence of this reaction could be explained by a tight transition state with a small

threshold energy. The pressure dependence can also be explained if a stable intermediate is formed.^{7,8}



A potential energy surface for such a reaction is shown in Figure 1. As shown in the first paper of this series,⁹ the presence of an excited intermediate can explain both strongly curved Arrhenius plots and negative temperature dependences in what otherwise appear to be elementary reactions. We have developed a modified version of RRKM theory for calculating rate constants for reactions of this type. The theory allows for the possibility that the formation of the intermediate may not be fast relative to the decomposition to products and permits V_2 to be less than V_1 . In this paper we apply this theory to the OH + CO reaction.

Smith and Zellner⁷ carried out transition-state theory calculations for a mechanism involving an HOCO intermediate. They obtained reasonably good agreement with experiment but found too strong a temperature dependence and too low a rate constant

(1) Baulch, D. L.; Drysdale, D. D. *Combust. Flame* 1974, 23, 215.

(2) DeMore, W. B.; Molina, M. J.; Watson, R. T.; Golden, D. M.; Hampson, R. F.; Kurylo, M. J.; Howard, C. J.; Ravishankara, A. R. Jet Propulsion Laboratory California Institute of Technology, Pasadena, CA, 1983, JPL Publ. 83-62.

(3) Perry, R. A.; Atkinson, R.; Pitts, J. N. *J. Chem. Phys.* 1977, 67, 5577.

(4) Paraskevopoulos, G.; Irwin, R. S. *J. Chem. Phys.* 1984, 80, 259.

(5) DeMore, W. B. *Int. J. Chem. Kinet.*, in press.

(6) Dryer, F.; Nageli, D.; Glassman, I. *Combust. Flame* 1971, 17, 270.

(7) Smith, I. W. M.; Zellner, R. *J. Chem. Soc., Faraday Trans. 2* 1973, 69, 1617.

(8) Smith, I. W. M. *Chem. Phys. Lett.* 1977, 49, 112.

(9) Mozurkewich, M.; Benson, S. W. *J. Phys. Chem.*, preceding paper in this issue.